

Little Red Dots: Super-Eddington Stellar Engines, Not Overmassive Black Holes

arXiv:2606.30711 — Little Red Dots as intermediate mass, super-Eddington engines.

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Type IIn supernovae and eta Carinae's Great Eruption offer a unified model for JWST's enigmatic Little Red Dots.

Background

Little Red Dots display a unique combination of features unseen in other galaxies or AGN. Their blackbody-like continua and broad lines challenge standard interpretations.

Are Little Red Dots overmassive black holes or something else? Their true nature remains unresolved.

What this paper is selling

The authors propose that dense winds from intermediate-mass engines create pseudo-photospheres, mimicking black holes. This explains the entire LRD life cycle, from enshrouded birth to massive black hole seeds.

Key results to remember

- LRDs share spectral features with Type IIn SNe and eta Carinae's Great Eruption.
- Engine masses are $<10^5$ solar masses, with super-Eddington luminosities.
- Dust production may mark the transition to classical AGN.

Fig. 4. LRDs, Type IIn SNe, and eta Carinae cluster at similar effective temperatures...

Fig. 8. Schematic: a cold wind enshrouds the engine, then hot winds shock, converting...